ROS 2 Quick Start

DR-xcobo 2.0 | v1.0 | ROS 2 Jazzy / RViz2 | English

Environment: Ubuntu 24.04 LTS Desktop with ROS 2 Jazzy Desktop already installed. Use a Bash terminal.

Download the model ZIP into **~/Downloads/** and keep its filename. Run the two blocks below in order, in the same terminal.

First use: install tools

```
# 1. Load ROS and install tools (once)
source /opt/ros/jazzy/setup.bash
sudo apt update
sudo apt install build-essential unzip python3-colcon-common-extensions \
  ros-jazzy-joint-state-publisher-gui
```

This installs model dependencies, not a complete ROS distribution. If the first line reports a missing file, install the ROS version above first.

First use: extract, build & launch

```
# 2. Extract the ZIP (once)
mkdir -p ~/ros2_ws/src
unzip ~/Downloads/dr-xcobo-2.0-ros2-urdf-v1.0.zip -d ~/ros2_ws/src

# 3. Build and open the model
cd ~/ros2_ws
colcon build --symlink-install --packages-select dr_xcobo_description
source install/setup.bash
ros2 launch dr_xcobo_description display.launch.py
```

Check for **~/ros2_ws/src/dr_xcobo_description/package.xml**. Keep the complete meshes folder. Skip the extraction command if you have already extracted the ZIP.

Use a dedicated ROS terminal and workspace; do not mix ROS 1 and ROS 2 environments. The launch opens the robot display and a separate joint-slider window.

Why source the setup files?

source loads ROS and package search paths into the current terminal. Usually, you only need it once per terminal. A new terminal needs the setup again, manually or automatically as shown on page 2. Sourcing does not install or rebuild software.

Package structure, launch settings, and mesh references were checked; no build or runtime test was performed in ROS. This bundle displays the model. Planning, physics, and hardware control require separate setup. Original effort and velocity limits are zero; review them before controller use.

Launch & Use

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Manual setup: three lines in each new terminal

After a successful build, open a fresh Bash terminal and run:

```
source /opt/ros/jazzy/setup.bash
source ~/ros2_ws/install/setup.bash
ros2 launch dr_xcobo_description display.launch.py
```

Line 1: load ROS. **Line 2:** load the model package. **Line 3:** open the model and sliders.

ROS 2 does not require roscore.

Optional: load the environment with .bashrc

After the first successful build, run **nano ~/.bashrc**. Add the following two lines at the end, once; do not duplicate existing entries:

```
source /opt/ros/jazzy/setup.bash
source ~/ros2_ws/install/setup.bash
```

Press **Ctrl+O**, **Enter** to save, then **Ctrl+X** to exit. Open a new terminal. It will load the environment automatically; you only need **line 3 of the startup block above** to launch the model.

Add only your chosen version; do not auto-load ROS 1 and ROS 2 together. Before switching versions or workspaces, remove or comment out (#) the old setup lines, open a new terminal, and source the target environment manually.

Before rebuilding: comment out the ~/ros2_ws/install/setup.bash line in .bashrc, then open a new terminal to build. Restore the line after a successful build and open a new terminal.

Basic controls

View: use Move Camera in RViz2 to orbit, pan, and zoom.
Joints: drag sliders in Joint State Publisher GUI.
Stop: press **Ctrl+C** in the launch terminal.

Optional: display a static pose without sliders

Stop the current launch with Ctrl+C, then run in the same terminal:

```
ros2 launch dr_xcobo_description display.launch.py gui:=false
```

If something goes wrong

Package not found: check the folder on page 1, complete the build, and run the three startup lines again.
Incomplete model: keep all mesh files and set Fixed Frame to **base_link**.
No sliders: install joint-state-publisher-gui and launch without gui:=false.

In RViz2, RobotModel uses Topic **/robot_description** with **Transient Local** durability.

Course resources | Model ZIP | ROS installation | Environment setup

ROS 2 Quick Start

DR-xcobo 2.0 | v1.0 | ROS 2 Jazzy / RViz2 | 中文版

环境：Ubuntu 24.04 LTS 桌面版，已安装 ROS 2 Jazzy Desktop。使用 Bash 终端。

下载模型 ZIP，将原文件名保留并放在 ~/Downloads/。下面两段代码在同一终端依次执行。

首次使用：安装工具

```
# 1. 加载 ROS 并安装工具（只需一次）
source /opt/ros/jazzy/setup.bash
sudo apt update
sudo apt install build-essential unzip python3-colcon-common-extensions \
  ros-jazzy-joint-state-publisher-gui
```

这一步安装模型所需工具，不会安装完整 ROS。若第一行提示文件不存在，请先安装上面的 ROS 版本。

首次使用：解压、编译与启动

```
# 2. 解压模型包（只需一次）
mkdir -p ~/ros2_ws/src
unzip ~/Downloads/dr-xcobo-2.0-ros2-urdf-v1.0.zip -d ~/ros2_ws/src

# 3. 编译并打开模型
cd ~/ros2_ws
colcon build --symlink-install --packages-select dr_xcobo_description
source install/setup.bash
ros2 launch dr_xcobo_description display.launch.py
```

解压后应存在 ~/ros2_ws/src/dr_xcobo_description/package.xml。保留完整 meshes 文件夹；已经解压过则跳过解压命令。

使用单独的 ROS 终端和工作空间，避免混用 ROS 1、ROS 2 环境。启动后会打开模型和关节滑块两个窗口。

为什么需要 source?

source 为当前终端加载 ROS 及功能包的查找路径。同一终端通常不用重复；新开终端需要重新加载，或按第 4 页设置自动加载。这一步不重新安装或编译软件。

已核对包结构、启动配置和网格引用，尚未在 ROS 环境中实际编译运行。本包用于模型显示；运动规划、物理仿真及真实机器人控制需另行配置。原始关节 effort 和 velocity 限制为 0，使用控制器前须核对。

日常启动与操作

DR-xcobo 2.0 | ROS 2 Jazzy / RViz2 | 中文版

手动加载：每个新终端运行三行

编译成功后，在新的 Bash 终端中执行：

```
source /opt/ros/jazzy/setup.bash
source ~/ros2_ws/install/setup.bash
ros2 launch dr_xcobo_description display.launch.py
```

第 1 行：加载 ROS；**第 2 行**：加载模型包；**第 3 行**：打开模型与滑块。

ROS 2 无需启动 roscore。

可选：用 .bashrc 自动加载

首次编译成功后，执行 nano ~/.bashrc，在文件末尾添加以下两行，已有则不重复添加：

```
source /opt/ros/jazzy/setup.bash
source ~/ros2_ws/install/setup.bash
```

按 Ctrl+O、Enter 保存，Ctrl+X 退出，再新开一个终端。以后系统会自动加载环境，直接运行上方三行中的**第 3 行启动命令**即可。

只添加对应版本的两行，不要同时自动加载 ROS 1 和 ROS 2。切换版本或工作空间前，先删除或注释（行首加 #）旧的自动加载行，再新开终端手动 source 目标环境。

再次编译前：先在 .bashrc 中注释加载 ~/ros2_ws/install/setup.bash 的一行，再新开终端编译。成功后恢复该行并重开终端。

最简单的操作

看模型：RViz2 中使用 Move Camera 工具旋转、平移、缩放。
动关节：在 Joint State Publisher GUI 中拖动滑块。
退出：在启动终端按 Ctrl+C。

可选：只显示模型，不打开滑块

先按 Ctrl+C 结束当前启动，再在同一终端执行：

```
ros2 launch dr_xcobo_description display.launch.py gui:=false
```

遇到问题时

找不到功能包：检查第 3 页目录结构，确认编译成功，再运行上面的三行。
模型显示不全：保留完整 meshes 文件夹，将 Fixed Frame 设为 base_link。
没有滑块：确认安装了 joint-state-publisher-gui，且启动时未加 gui:=false。

RViz2 中 RobotModel 使用 Topic：/robot_description；Durability Policy 为 Transient Local。

课程资源 | 模型 ZIP | ROS 安装说明 | 环境配置说明